

Reproducing and Extending Nash Regret Experiments in Zero-Sum Matrix Games

Project Report

Based on the paper

*On the Limitations and Possibilities of Nash Regret Minimization
in Zero-Sum Matrix Games Under Noisy Feedback*

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<https://github.com/MateoLopez00/Zero-Sum-MatrixGames.git>

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Abstract

This report reproduces and extends the empirical study of Nash regret minimisation in zero-sum matrix games under noisy feedback by Maiti, Jamieson and Ratliff [MJR23]. We reproduce the paper’s full-information diagonal-game experiments and its 2×2 bandit experiments, then study bilateral learning, Gaussian noise robustness, and a beyond- 2×2 bandit stress test. The reproduced settings show the expected flatter regret growth for the proposed algorithms relative to generic baselines. The extensions identify which assumptions matter: noise-aware delays help in the full-information game, the original bandit algorithm remains the relevant noisy Section 4 comparison, bilateral learning changes the opponent model, and larger bandit games expose limits outside the proved 2×2 regime. The report states horizons, seeds, run counts, game sizes, adversaries, and noise levels to support reproducibility.

1 Introduction

Nash regret minimisation in zero-sum matrix games studies whether a learner can maintain payoffs close to the value of the game while facing a strategic or adversarial column player. The source paper [MJR23] shows that this is possible in selected feedback models, but also that generic bandit or online-learning methods can fail under adversarial play.

Our project has five contributions: reproducing Section 3 full-information experiments; reproducing Section 4 bandit experiments; studying bilateral learning; testing Gaussian noise robustness; and applying the Section 4 bandit construction to larger diagonal games outside the proved 2×2 regime.

The report is organized as follows. Section 2 defines games, feedback models, algorithms, adversaries, and the regret metric. Section 3 gives the exact experimental settings and reproduction results. Sections 4–6 present the three extensions. Section 7 discusses reproducibility, limitations, and the conclusions supported by the experiments.

2 Models, Metrics, and Assumptions

2.1 Zero-sum matrix games and Nash regret

Let $A \in [0, 1]^{m \times n}$ be the row player’s payoff matrix. For mixed strategies x and y , the expected payoff is $x^\top Ay$, and the value of the zero-sum game is

$$V^* = \max_x \min_y x^\top Ay.$$

At round t , the row learner plays x_t and the column player chooses y_t according to a fixed rule or adversary. We report cumulative Nash regret

$$\text{Reg}_T = \sum_{t=1}^T \left(V^* - x_t^\top Ay_t \right),$$

with the sign convention used in the implementation: smaller cumulative regret indicates that the row learner remains closer to the game value.

2.2 Feedback models

The project uses three feedback models. In the Section 3 full-information setting, the learner observes a full payoff signal on a diagonal game with $A_{ii} = 0.4 + 0.2(i - 1)/(n - 1)$ and $n \in \{10, 20, 50, 100\}$. In the Section 4 bandit setting, the learner observes only the played cell in the 2×2 game

$$A = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad V^* = 2/9.$$

In the Gaussian-noise extension, a full-information observation is clipped to $[0, 1]$ after adding Gaussian noise entrywise, while the bandit version observes only

$$\text{clip}(A_{i_t j_t} + \sigma Z_t, 0, 1), \quad Z_t \sim \mathcal{N}(0, 1).$$

This clipping assumption keeps payoffs bounded but can bias observations near 0 and 1; therefore noisy-extension curves are not expected to match noiseless curves numerically.

2.3 Algorithms and baselines

The full-information experiments compare an empirical Nash baseline, Hedge, and the paper’s proposed algorithm. Hedge is a standard expert-advice method [FS97, CBL06]. The bandit experiments compare UCB, EXP3, and OURALG. UCB is a stochastic-bandit baseline [ACBF02], while EXP3 is the standard adversarial-bandit baseline [ACBFS02]. The noise-aware full-information variant modifies only the update delay of the proposed method; the Section 4 noise experiments keep the original bandit OURALG rather than adding a separate noise-aware variant.

2.4 Adversaries and assumptions

Each Section 4 trial has two phases of length $T/2$. Phase 1 uses one of the three adversaries from the paper: threshold best response, tolerance-band best response, or fixed Nash play $(1/3, 2/3)$. Phase 2 always uses pure best response. The Section 4 noise extension keeps exactly these adversaries and phases; only the observation channel changes. All results are finite-horizon Monte Carlo estimates rather than asymptotic proofs.

3 Reproduction Methodology and Results

Figure 1 summarises the two reproduction experiments using cumulative regret curves. Table 1 gives the exact settings used for the report figures and tables. Short source identifiers are used in the table to avoid unreadable path overflow; the identifiers refer to the notebooks and scripts listed in the final paragraph of this section. Figures are copied into `Final_report/figures` only to make the LaTeX package self-contained; the experiments themselves are regenerated from the original notebooks and scripts.

The reproduction is technically consistent with the paper’s assumptions. In the full-information setting, the strongest qualitative signal is not a single terminal value but the curve shape: the proposed algorithm pays an early exploration cost and then slows, while the baselines keep accumulating regret more steadily, especially for smaller diagonal games. In the bandit setting, the adversarial phase structure is crucial. Adversary 3 is benign in Phase 1 because it plays Nash, but the Phase 2 best response sharply separates algorithms. This matches the theoretical motivation: generic UCB and EXP3 are not designed for this adversarial game protocol, while OURALG uses the structure of the 2×2 matrix game.

For provenance, S3 curves are generated by the Section 3 Nash-regret curves notebook, S4 curves by the Section 4 Nash-regret curves notebook, S3/S4 noise by the two noise-robustness

Experiment	Reported settings	Source
Section 3 reproduction	Diagonal games $n \in \{10, 20, 50, 100\}$; cumulative Nash regret; $T = 100,000$; six seeds; algorithms: Nash, Hedge, Our-Algo.	S3 curves
Section 4 reproduction	2×2 game with $V^* = 2/9$; adversaries 1–3; two phases of $T/2$; $T = 100,000$; 32 seeds; algorithms: UCB, EXP3, OurAlg.	S4 curves
Section 3 noise	Full-information Gaussian-clipped feedback; $n = 100$ convergence plots; $\sigma \in \{0.1, 0.2, 0.3\}$; $T = 100,000$; 12 runs; seed 7.	S3 noise
Section 4 noise	Same Section 4 adversaries and phases; played-cell Gaussian-clipped feedback; $\sigma \in \{0.1, 0.2, 0.3\}$; $T = 100,000$; 32 runs; seed 42.	S4 noise
Bilateral learning	Both players update rather than fixing the column rule; diagonal full-information game shown at $n = 30$; 2×2 bandit game shown with bilateral learner pairs only.	Bilateral
Beyond 2×2	Diagonal bandit games $n \in \{2, 3, 4, 5, 7, 10\}$; T up to 10^6 ; 128 trials; algorithms: UCB, EXP3, OurAlg.	S4 larger- n

Table 1: Reproducibility settings used in the report. These are scientific settings, not internal code preset labels.

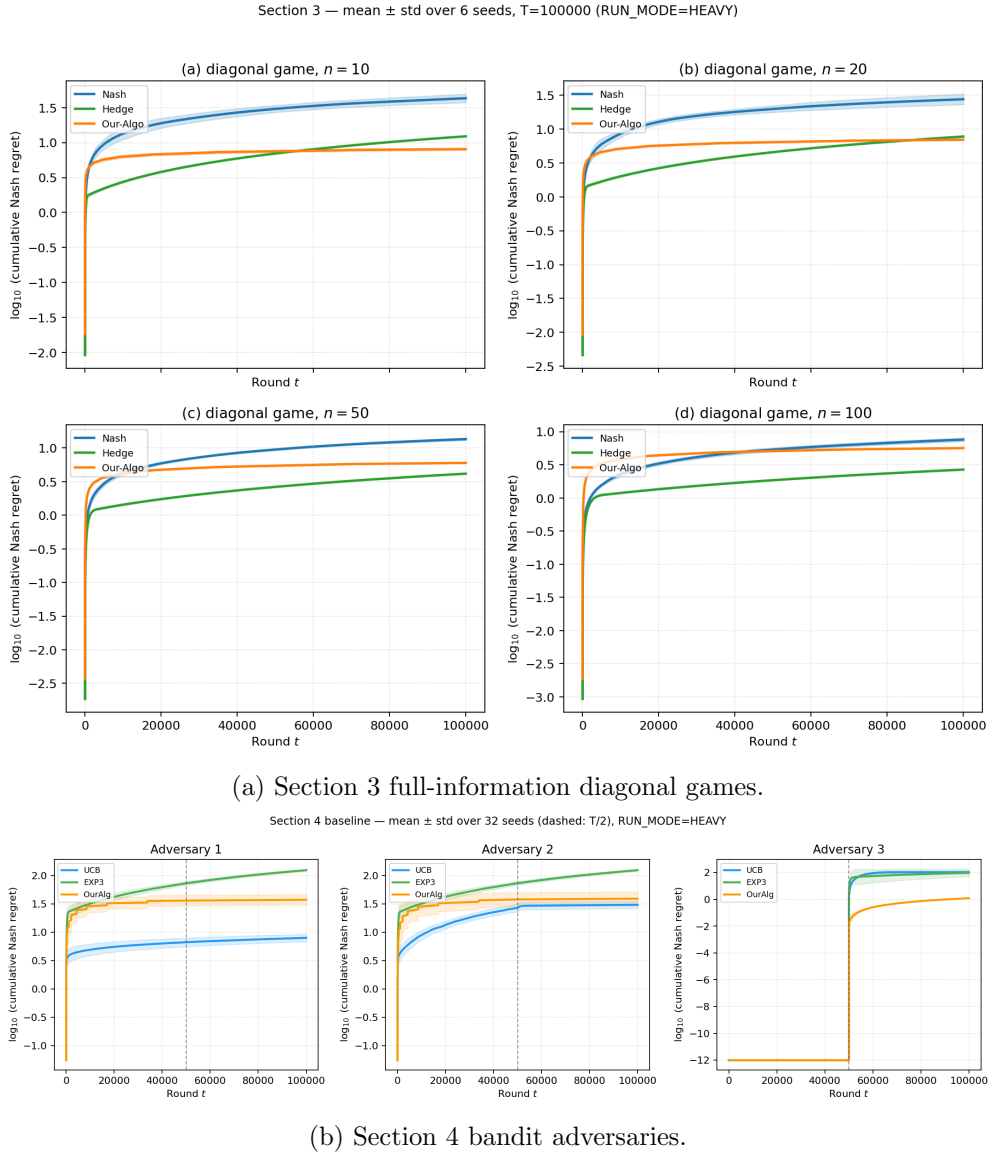


Figure 1: Reproduction cumulative Nash-regret curves at $T = 100,000$. Panel (a) shows the Section 3 full-information diagonal games; panel (b) shows the Section 4 two-phase bandit adversaries.

notebooks, S4 larger- n by the beyond- 2×2 notebook, and bilateral learning by the two bilateral scripts. These source names are intentionally summarized here; the repository structure contains the exact files and copied figures.

4 Noise Robustness Extension

The main extension tests a controlled change in the feedback model. For Section 3, we add Gaussian noise to the full-matrix feedback and clip entries into $[0, 1]$. For Section 4, only the played bandit cell is observed as

$$\text{clip}(A_{i_t j_t} + \sigma Z_t, 0, 1), \quad Z_t \sim \mathcal{N}(0, 1).$$

Clipping is an important modelling assumption: it keeps payoffs in range but can bias observations near 0 and 1. Therefore the noisy-extension curves should not be expected to numerically reproduce the noiseless Section 4 curves.

Full-information noise-aware variant. The baselines remain unchanged. The extension modifies only the paper’s algorithm by delaying updates more when feedback is noisier:

$$\tau_{\text{orig}} = \min\{\log^2(T), \sqrt{T}\}, \quad \tau_{\text{noise}} = \min\{(1 + 2\sigma) \log^2(T), \sqrt{T}\}.$$

The experiment uses $T = 100,000$, $n = 100$ for the convergence plots, 12 independent runs, seed 7, and $\sigma \in \{0.1, 0.2, 0.3\}$. Table 2 reports terminal regret at $\sigma = 0.3$ over the diagonal games used in the sweep.

n	Nash	Hedge	Our-Algo	Noise-aware	Reduction
10	113.08	16.30	8.54	8.25	3.4%
20	58.66	15.48	6.36	6.08	4.4%
50	24.41	15.33	5.02	4.78	4.8%
100	12.99	15.75	4.58	4.30	6.1%

Table 2: Section 3 noise robustness at $\sigma = 0.3$, $T = 100,000$, 12 runs. The reduction is relative to the original Our-Algo.

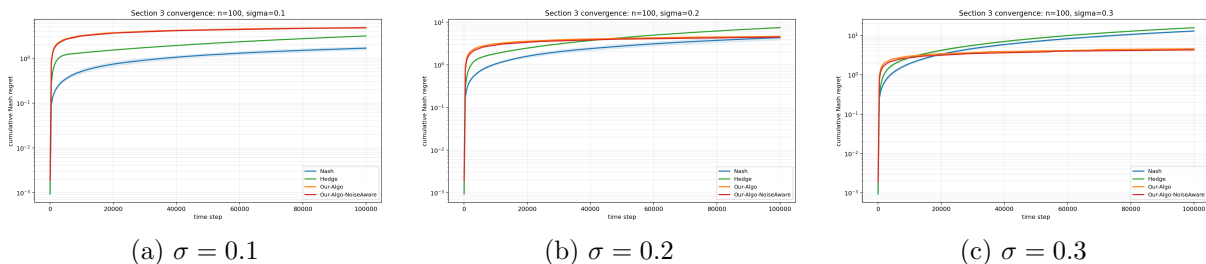


Figure 2: Section 3 noise convergence for $n = 100$ and $T = 100,000$. The noise-aware variant becomes more useful as σ increases.

The result supports the intended mechanism. When σ is small, the empirical matrix is reliable enough that the extra delay has little effect, and Nash can still be competitive at $n = 100$. At higher noise, premature updates based on noisy estimates become more costly, while the two proposed variants flatten relative to Nash and Hedge. The improvement from the noise-aware threshold is moderate rather than dramatic, which is also an important caveat: the original method is already robust in this setting.

Bandit noise with the same adversaries. For Section 4, we do not introduce a noise-aware bandit variant in the final comparison. The adversaries and two-phase protocol are exactly the same as in the reproduction: the extension imports the same adversary logic from `section4.bandit.py`. The only change is the feedback distribution, from Bernoulli observations to clipped Gaussian played-cell observations. The reported runs use $T = 100,000$, 32 independent runs, seed 42, and $\sigma \in \{0.1, 0.2, 0.3\}$.

Adversary	UCB	EXP3	OurAlg
1	7.53	141.27	77.90
2	59.01	140.74	86.07
3	778.26	703.05	2.86

Table 3: Final Section 4 Nash regret at $\sigma = 0.3$, $T = 100,000$, 32 runs.

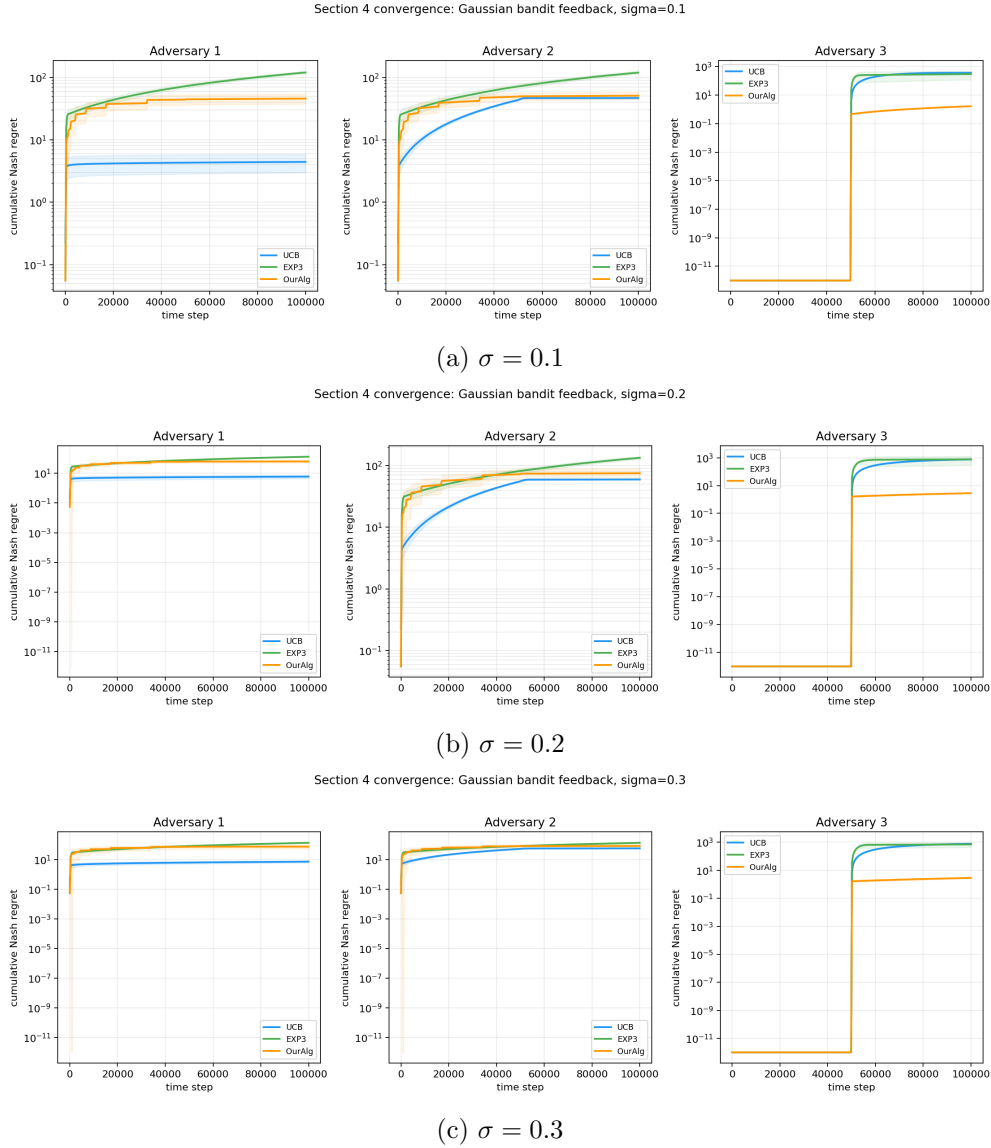


Figure 3: Section 4 noisy bandit convergence, $T = 100,000$. Each row contains Adversaries 1–3 for one noise level.

The bandit extension should be interpreted relative to its modified feedback channel. Since clipped Gaussian samples have different variance and bias properties from Bernoulli observations,

rankings need not match the noiseless Figure 2 reproduction. The most stable qualitative finding is on Adversary 3: at high noise, UCB and EXP3 accumulate very large regret after the phase switch, whereas OURALG remains close to the game value. Against Adversaries 1 and 2, rankings are more adversary-dependent: UCB can remain competitive, EXP3 is usually the largest-regret baseline, and OURALG sits between them in these noisy finite-horizon runs.

5 Bilateral Learning Extension

The paper evaluates a row learner against a fixed or adversarial column rule. Our bilateral extension changes this setting by allowing both players to update their strategies simultaneously from feedback. This constitutes a different learning scenario rather than a simple parameter variation, since the standard best-response regret interpretation no longer directly applies in self-play. In particular, the instantaneous payoff $x_t^\top A y_t$ may exceed the equilibrium value V^* , making the signed equilibrium gap negative.

Instead of reporting signed regret, we measure the cumulative absolute equilibrium gap

$$\sum_{t=1}^T |V^* - x_t^\top A y_t|,$$

which serves as a stability metric indicating whether the joint dynamics converge toward the Nash equilibrium value over time.

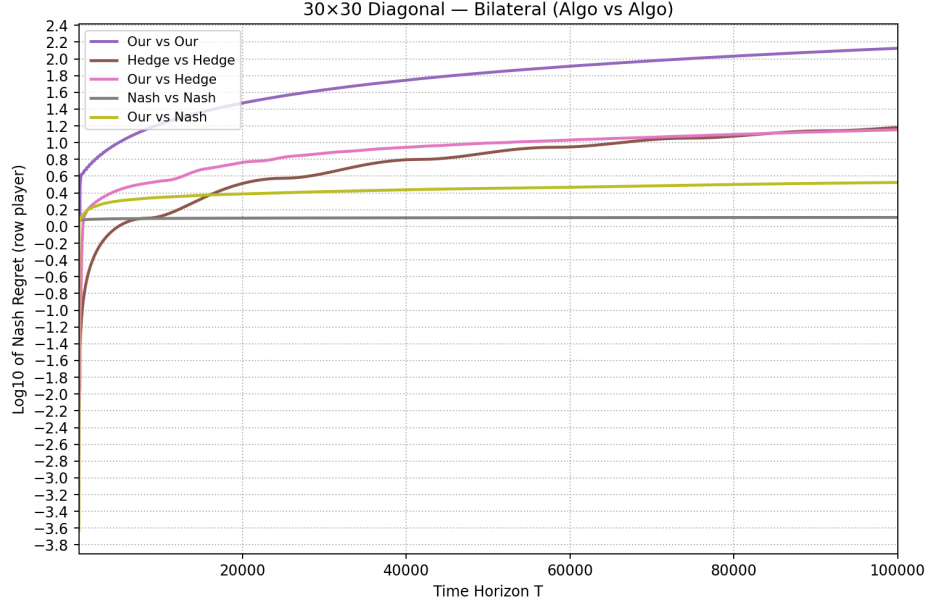
In the full-information diagonal game, both players observe sampled matrix entries and run pairings such as OURALG–OURALG, Hedge–Hedge, Nash–Nash, and mixed pairs. In the 2×2 bandit game, both players use bandit-feedback updates on the same matrix game as in the Section 4 reproduction.

The results should not be interpreted as contradicting the original paper when they differ from the adversarial baseline, since the opponent model changes fundamentally in the bilateral setting.

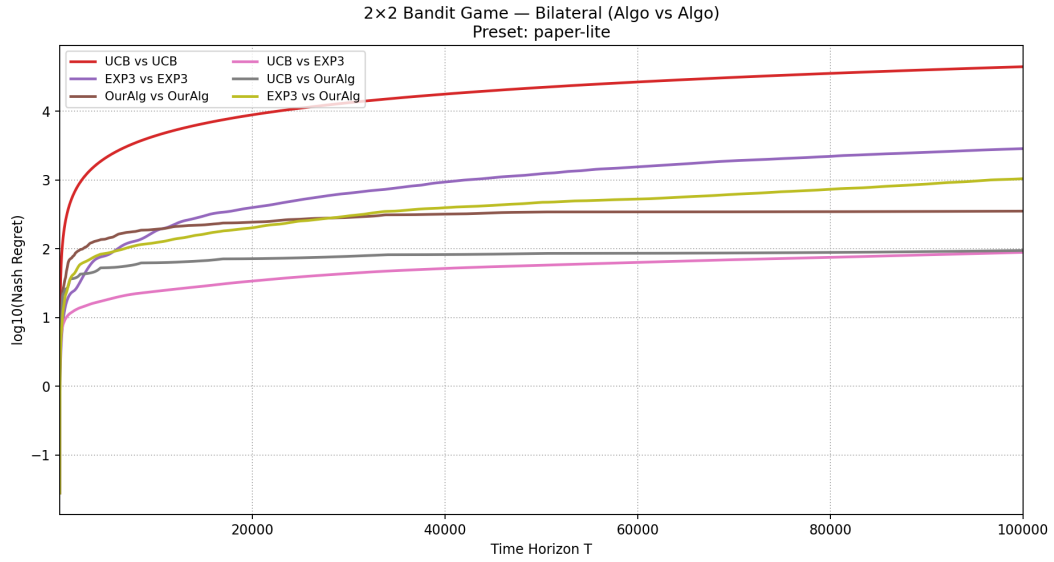
In the Section 3 experiment, Nash–Nash remains nearly flat because both players repeatedly play the empirical Nash equilibrium of their current estimates, which keeps the observed payoff close to V^* from early rounds onward and avoids large exploratory deviations. In contrast, OURALG–OURALG exhibits the largest cumulative absolute equilibrium gap because OURALG includes an aggressive warm-up and reinitialization phase. When both players simultaneously explore in this manner, the joint dynamics initially drift farther from V^* before eventually stabilizing. Mixed pairings such as OURALG–Nash and OURALG–Hedge lie between these extremes, reflecting the interaction between an exploratory row learner and a more stable column strategy.

In Section 4, UCB–UCB produces the largest growth because UCB was originally designed for single-agent stochastic optimization rather than simultaneous equilibrium learning. The optimistic updates of the row player and pessimistic updates of the column player can induce persistent oscillatory behavior, delaying convergence toward the Nash value. EXP3–EXP3 also accumulates a large equilibrium gap because EXP3 maintains persistent exploration throughout the horizon. By comparison, OURALG–OURALG plateaus at a lower level than both baselines, suggesting that its update structure is more compatible with bilateral convergence in the 2×2 setting.

Overall, these experiments indicate that bilateral learning introduces a distinct dynamical regime compared with the original adversarial formulation. The setting is therefore useful for studying adaptive multi-agent interaction, although it falls outside the assumptions of the original regret analysis, which considers a fixed adversarial column player rather than a simultaneously adapting opponent.



(a) Section 3 diagonal game, $n = 30$.



(b) Section 4 bandit game.

Figure 4: Bilateral-learning extension. Both panels show learner-vs-learner curves only: full-information diagonal-game pairings in panel (a), and 2×2 bandit pairings in panel (b).

6 Non-Adversarial Column Player Extension

The algorithm by Arnab et al. minimizes the regret against an adversarial column player, but how does the algorithm react when the column player is not deliberately trying to minimize the row player’s return? This extension explores how the algorithm from Arnab et al. adapts to less hostile and presumably easier environments.

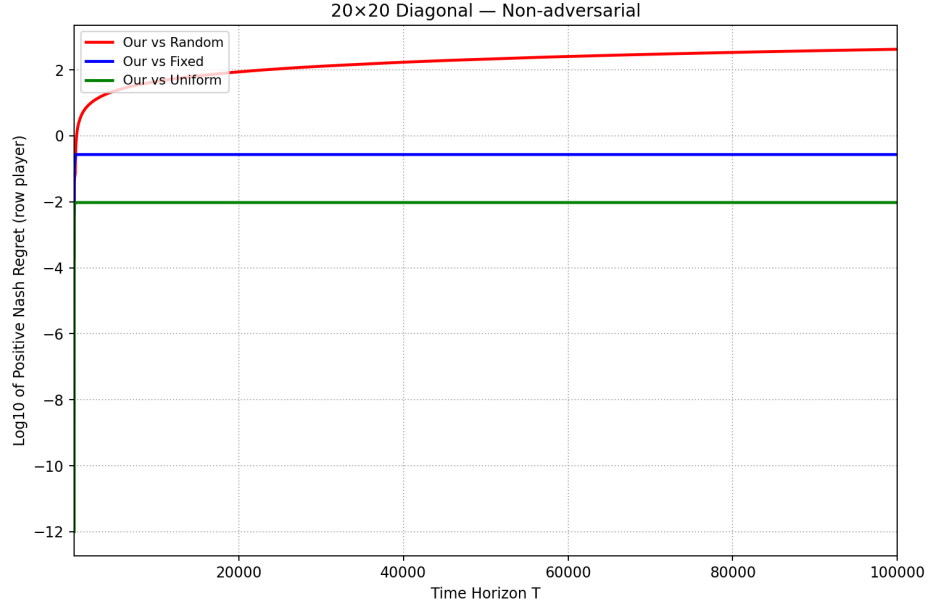
We use three non-adversarial responses, a fixed response, a uniform response, and a random response. A fixed response is a response where the column player plays the same action in every round. Uniform response means the column player always plays the strategy that gives equal probability to each action, specifically $y_t = \frac{1}{m}\mathbf{1}$, $\forall t \in T$. For the random response you draw a new, random simplex in each round using the Dirichlet distribution, which will then serve as the column player’s strategy for that round.

The algorithm from the paper consistently incurs negative Nash regret against all three non-adversarial column players, but from time to time the game value of a round is below the Nash equilibrium’s game value. We chose the metric to capture only when the algorithm underperforms the game value in the Nash equilibrium. The metric is the accumulated positive Nash regret. Rounds with a negative Nash regret are ignored.

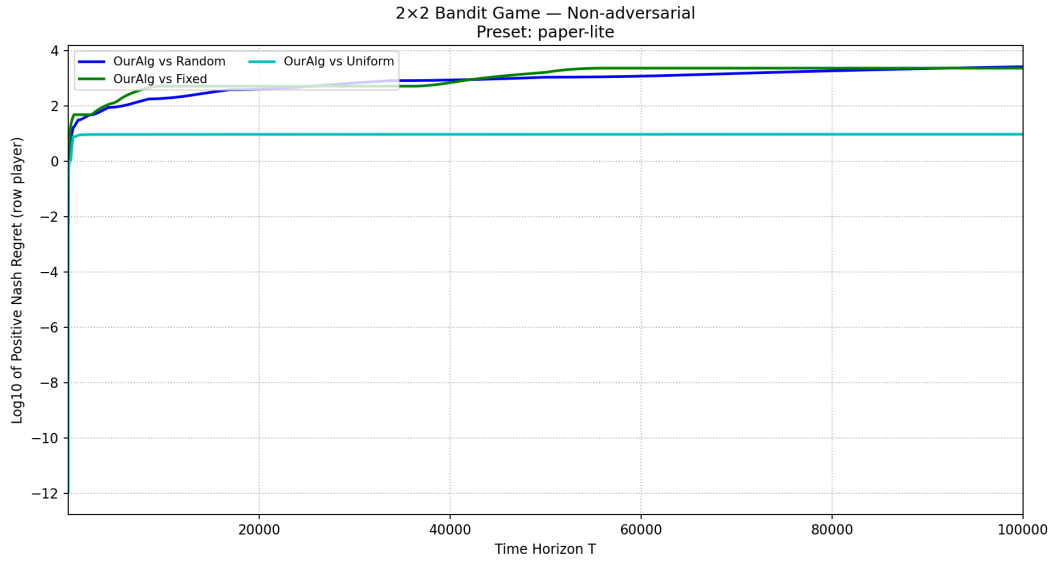
In the full information setting the row player quickly learns how to play against a fixed response from the column player or a random response, see Fig. fig:non-adversarial. After a short warm-up period the algorithm from Maiti et al. never underperforms the game value in the Nash equilibrium ever again. However, the algorithm struggles when the column player responds with a random strategy. The learned knowledge is of little value and avoiding positive Nash regret can not be guaranteed.

In the Bandit setting, the row player first needs to explore all entries of the matrix. He fails to do that against a column player, that only plays one fixed column. The algorithm from Maiti et al. does not account for that behavior and the row player regularly incurs positive Nash regret.

All that said, if you take all rounds into account, the incurred Nash regret is negative against all non-adversarial responses in both settings. This affirms the paper’s claim that their algorithm achieves below $\sqrt{T}$ Nash regret against any column player.



(a)



(b)

Figure 5: Row player's accumulated positive Nash regret when playing against a non-adversarial column player: results from full-information setting in panel (a), and results from 2×2 bandit setting in panel (b).

7 Bandit Nash Regret Beyond 2×2 : Diagnostic Extension

We applied the paper’s Algorithm 6 to larger diagonal bandit games ($n \in \{2, 3, 4, 5, 7, 10\}$) with 128 trials and horizons up to $T = 10^6$. The extension generalizes all algorithmic components like gradient, strategy update, phase structure, and epoch logic to their natural n -dimensional analogues. However, two implementation choices optimize for the diagonal game class: the Nash equilibrium and the D parameter are computed from diagonal entries of the UCB matrix F_2 only, rather than from the full matrix. This is a computational choice for stability and speed, not a mathematical requirement.

n	UCB	EXP3	OurAlg	OurAlg slope
2	27.38	340.27	12.30	0.31
3	37.19	890.49	21.33	0.45
4	41.05	1557.19	40.47	0.73
5	43.14	2294.43	98.32	0.73
7	45.07	3882.14	609.88	1.04
10	45.91	6449.91	6993.62	1.07

Table 4: Table 4 reports terminal Nash regret at $T = 10^6$ and terminal (final decade) log-log slope. The terminal slope shows the limiting growth rate as $T \rightarrow \infty$, while the absolute regret comparison against UCB shows the practical performance boundary.

Why the diagonal optimization matters. The paper’s Algorithm 6 computes Nash using all entries of the full UCB matrix $F_2 = \hat{B} + \text{bonus}$, including off-diagonal entries which carry confidence bonuses from unobserved pairs. For general matrices, these off-diagonal estimates are signal. For diagonal games, they are pure noise and the true game has zero off-diagonal entries, so large off-diagonal bonuses distort Nash estimation during exploration. By computing Nash from diagonal entries only, we filter this noise and provide the algorithm with cleaner strategy direction. This is valid for the diagonal game class but does not transfer to general games. A purist port of Algorithm 6 to general $n \times n$ games would use full-matrix Nash, lose to UCB earlier than $n = 5$, and directly address the paper’s open problem. Our approach instead shows where failure occurs even under ideal conditions: exploration coverage degrades from $n = 5$ onward, becomes severe by $n = 10$, and prevents extension beyond the proved 2×2 regime.

8 Reproducibility and Limitations

Several assumptions should qualify the conclusions. The report is designed to be reconstructable without needing to inspect the implementation code. The payoff matrices, feedback models, adversary rules, algorithmic baselines, proposed variants, horizons, seeds, numbers of independent runs, game sizes, noise levels, and evaluation metrics are stated in the methodology and extension sections, especially in Table 1. The repository is included as supplementary material for exact execution and figure regeneration, not as a replacement for the methodological description.

First, all results are finite horizon Monte Carlo estimates. Shaded bands in the figures summarize empirical variation across independent runs using standard deviation style summaries where available; we did not bootstrap formal confidence intervals, so the bands should not be read as theorem level confidence statements. Second, no algorithm was selected by tuning on a held out validation problem: the report shows the paper algorithms and standard baselines under fixed settings. Third, Gaussian clipping changes the observation distribution; this is a reasonable bounded payoff model, but it can introduce bias near the boundaries. Finally, copied figures are used only for stable compilation, while the reported settings in the text are sufficient to understand and reconstruct the experiments.

Overall, the reproduction supports the paper’s main empirical claims: the proposed algo-

rithms show flatter regret growth than generic online learning baselines in the settings where the theory applies. The extensions show how sensitive that story is to assumptions: explicit noise awareness helps in the full information diagonal game, the original bandit algorithm remains the correct comparison under Section 4’s adversaries with noisy observations, bilateral learning changes the opponent model and produces genuinely different dynamics, and larger bandit games expose limitations outside the proved 2×2 regime.

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